

REVISED SYLLABUS

MEDICAL LAB TECHNIQUE (MLT)

Minor from Vocational

OBJECTIVE:

1. To identify the structures in the human body in their normal position.
2. To demonstrate the knowledge about each system with functional co-relation.
3. To develop the skills to use the knowledge of structures in the human body in their respective fields of function, with utmost care to the needs of the patients and with regards to the betterment of the health of the patients.
4. To develop and demonstrate the basic skills in microscopy of any given tissue, with an understanding of handling different types of microscopes independently in the course of Laboratory technology.

SEM-II

(PAPER-01 Credit-04 F.M. 100)

THEORY:

FM: 60+15= 75

GENERAL ANATOMY OF HUMAN

1. General Anatomy
 - Digestive System
 - Respiratory System
 - Excretory System
 - Circulatory System
 - Reproductive System
 - Nervous System
 - Lymphatic System
2. Anatomical positions (Through models, charts, pictures and digital modes)
 - Supine
 - Prone

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3. Different parts of human body
 - Axial and Appendicular
 - Head and Neck
 - Thorax and Abdomen
 - Upper and Lower Limbs
4. Anatomical planes and sections
 - Median
 - Sagittal
 - Transverse
 - Longitudinal
 - Horizontal
 - Oblique
5. Anatomical terms (brief)
 - Anterior and Posterior
 - Superior and Inferior
 - Medial and Lateral
 - Proximal and Distal
 - Superficial and Deep
 - Ventral and Dorsal
 - Cephalic and Caudal
 - Interior and Exterior
 - Invagination and Evagination

PRACTICAL

FM: 25

1. Anatomical positions (through models, charts, pictures and digital modes)
2. Location and identification of Limbs: Upper and Lower limb
3. Location and identification of Heart, Trachea and Lungs
4. Location and identification of organs in GIT and its associated structures like liver and gall bladder, pancreas.
5. Urinary system: Identification of parts and functional co-relation.
6. Reproductive system: Identification of parts with functional co-relation.
7. CNS: Meninges and parts of CNS, Identification with functional co-relation.

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THEORY: FM: 60+15= 75
HUMAN PHYSIOLOGY AND ENDOCRINOLOGY:

1. Digestive system:
 - GIT Associated glands and its secretion
 - Disorder (Diarrhea)
2. Cardiovascular system:
 - Structure of Heart
 - Blood
3. Respiratory system:
 - Respirator organs
 - Mechanism of breathing and its control
 - Disorder (Heart attack/ Cardiac arrest)
4. Excretory system:
 - Excretory organs
 - Urine formation and absorption
 - Disorders (Diabetes mellites, Diabetes insipidus)
5. Nervous system
 - CNS, PNS, ANS
 - Cerebrospinal fluid
6. Reproductive system
 - Structure of Gonads and its associated glands
 - Primary and secondary sexual characters
 - Disorder (Infertility)
7. Lymphatic system
 - Primary lymphoid organs and their function
 - Secondary lymphoid organs and their function
8. Endocrine system
 - Secretions and functions of different endocrine glands: Pituitary, Islet of Langerhans, Thyroid
 - Endocrine disorders

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PRACTICAL

FM: 25

1. Collection of blood samples and anticoagulants and preparation
 2. Preparation of blood smear
 3. Determination of RBC, WBC, PLATELET
 4. Differential leucocytes count
 5. ESR, PCV (Demonstrate)
 6. Blood grouping and Rh typing
 7. Estimation of hemoglobin percentage
 8. Determination of blood pressure in man through Sphygmomanometer
 9. Determination of temperature, drugs, hormones, exercise on the rate of heart beat
 10. Kidney function tests: Blood urea estimation, Serum creatinine estimation, Serum uric acid
 11. Lipid profile tests: Serum Glucose, Total Cholesterol, HDL Cholesterol, Triglyceride
 12. Liver function tests: SGOT (AST), SGPT (ALT)
 13. Hormone assay: T3, T4, TSH
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SEM-VI
(PAPER-03 Credit-04 F.M. 100)

THEORY:
CYTOLOGY AND OESTEOLOGY:

FM: 60+15= 75

1. Animal cell and its organelles with their functions
2. Tissue and its types with their functions
3. Muscle contraction
 - Types of muscle and general muscle contraction
 - Types of muscle contraction: Concentric, Eccentric, Isocentric
4. Nerve conduction
 - Continuous
 - Saltatory
5. Bone and its type
 - Long
 - Short
 - Flat
 - Sesamoid
6. Bone joints
 - Fibrous
 - Cartilaginous
 - Synovial
7. Cartilage

PRACTICAL

FM: 25

1. Study of Permanent slides:
 - typical animal cell
 - types of animal tissue
 2. Study of bones through models, pictures, charts and digital mode
 - Bones: skull, vertebrae, pectoral, pelvic, limb
 - Bone joints: fibrous, cartilaginous, synovial
 3. Study through models, pictures, charts and digital mode
 - Muscle contraction
 - Nerve conduction
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SEM-VIII
(PAPER-04 Credit-04 F.M. 100)

THEORY:
TOOLS AND TECHNIQUES:

FM: 60+15= 75

1. Working principle, parts, applications, care and maintenance of instruments used in laboratory:
 - Water bath
 - pH meter
 - Laboratory balances
 - Incubators
 - Refrigerator
 - Colorimeter
 - Spectrophotometer
2. Centrifuge: Definition, Principle, Svedberg unit, centrifugal force, centrifugal field, rpm., types of centrifuges, use, care and maintenance of a centrifuge.
3. ELISA: Definition, Principle, types and application
4. PCR: Definition, Principle, types and application
5. Microscopy: magnification, resolution, types of microscopes (simple, compound, electron), focusing, care & handling of microscopes
6. Histology:
 - Fixatives used in histology, preparation, advantages & disadvantages
 - Stains used in histology, preparations and applications
 - Collection of tissue specimens, numbering and labelling
 - Tissue processing: fixation, dehydration, clearing, embedding
 - Microtomy: tissue sectioning, spreading
 - Staining procedure: Hematoxylin & Eosin (general), PAS (carbohydrate), Mercury, Ninhydrin- Schiff (protein), Sudan Black-B (lipid), Sudan III (lipid), Sudan IV (lipid), Methyl Green Pyronin-Y (DNA & RNA), Feulgen Reaction (RNA).

PRACTICAL

FM: 25

1. Practical problems during handling the microscope (loose mirror, loose eye piece, dried oil, etc) and histological techniques (artifacts, shrinkage, precipitate, Folds, Pinched tissue, Nick in knife, Autolysis, under and over staining).
2. Preparations of fixatives for tissue fixation
3. Preparations of stains for histological staining

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4. Tissue block (paraffin) preparation for histological studies
 5. Tissue processing, fixation, dehydration, clearing, embedding, tissue sectioning and H&E staining.
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COURSE OUTCOME:

Students will develop a preliminary idea of understanding body anatomy and histology of human body as well as appropriate vocabulary of appropriate terminology of various parts and organs. This will lay a strong foundation for advance study in higher semester.

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